

9 PROBABILITY DISTRIBUTIONS

(WMS: Chapter 2-4)

9.1 Probability of Events

We talk about the probability of an event, which is the outcome of an experiment. The sample space consists of all simple events which cannot be decomposed into simpler events. An event, E , is a subset of a sample space, S .

The **probability** $P(E)$ of an event E is a number representing the chance that the event occurs. It is defined such that

- $0 \leq P(E) \leq 1$
- $P(S) = 1$
- $P(\overline{E}) = 1 - P(E)$, where $\overline{E}$ (or E^c) is the complement of E
- If E_i, E_j are pairwise mutually exclusive, so $E_i \cap E_j = \emptyset$ (for $i \neq j$), then

$$P(E_1 \cup E_2 \cdots) = \sum_{i=1}^{\infty} P(E_i).$$

Conditional Probability

For events A, B , we define:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \text{ if } P(B) > 0$$

$P(A|B)$ represents the probability of A given B ; that is, the relative frequency of A given B . So

$$P(A \cap B) = P(A|B) \cdot P(B)$$

Independence

Events A, B are independent if any one of the following holds:

- $P(A|B) = P(A)$; that is, knowing B has happened doesn't tell you anything about the likelihood of A happening
- $P(B|A) = P(B)$; that is, knowing A has happened doesn't tell you anything about the likelihood of B happening
- $P(A \cap B) = P(A) \cdot P(B)$; that is, the same fraction of A s happen with B as in whole population

Ex. $P(\text{death in auto accident})$ and $P(\text{death auto accident} \mid \text{wearing seat belt})$ are not equal. These events (wearing seat belt and death in auto accident) are not independent.

Ex. To investigate racial profiling, which of the following do you want to compare: $P(\text{White}|\text{Frisk})$, $P(\text{Black}|\text{Frisk})$, $P(\text{Frisk}|\text{White})$, $P(\text{Frisk}|\text{Black})$?

Solution: We want to compare the probability of being frisked for Whites and Blacks; that is $P(\text{Frisk}|\text{White})$, $P(\text{Frisk}|\text{Black})$.

9.2 Random Variables and Probability Distributions

A **random variable**, Y , is a real-valued function whose domain is the sample space. In other words, Y is a function whose numerical value depends on outcome of an experiment.

There are two types of random variable. A **discrete** random variable takes on discrete or countably infinite values. Otherwise, Y is a **continuous** random variable.

ex. Number of days Dow Jones average is above a specific value in the current year: Discrete.

ex. Length of time Dow Jones average is above a specific value in the current year: Continuous.

To represent the probabilities of all the possible values of Y , we use a **distribution**. Distributions are defined differently for discrete and continuous variables so we treat them separately.

9.2.1 For Discrete Random Variable

Probability Distribution Function (p.d.f.), $p(y)$, defined by

$$p(y) = P(Y = y).$$

See Figure 1, showing the relative frequency of y . Properties of p :

- $0 \leq p(y) \leq 1$
- $\sum_y p(y) = 1$.

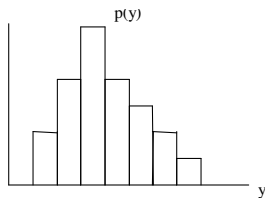


Figure 1: Horizontal axis is y -values. Vertical axis is relative frequency

Cumulative Distribution Function (c.d.f.), $F(y)$ defined by

$$F(y) = P(Y \leq y)$$

See Figure 2. Properties of F :

- $\lim_{y \rightarrow -\infty} F(y) = 0, \lim_{y \rightarrow \infty} F(y) = 1$,
- $F(y)$ is increasing so $y_1 < y_2 \Rightarrow F(y_1) \leq F(y_2)$.

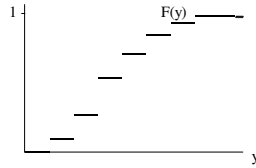


Figure 2:

ex. For three tosses of a fair coin, let Y be the total number of heads. The p.d.f. and c.d.f. of Y are:

y	0	1	2	3
$p(y)$	1/8	3/8	3/8	1/8
$F(y)$	1/8	4/8	7/8	1

For a Continuous Random Variable:

If Y can take an uncountable number of values, then the probability of getting any specific value, y , is zero; that is $P(Y = y) = 0$. Thus we can't use the same kind of probability distribution function as we used for a discrete random variable. However, the cumulative distribution function is defined as before, so we start with this.

Cumulative Distribution Function (c.d.f.), F , is defined by

$F(y) = P(Y \leq y)$ gives probability that $Y \leq y$. See Figure 3. Properties of F :

- $\lim_{y \rightarrow -\infty} F(y) = 0, \lim_{y \rightarrow \infty} F(y) = 1,$
- $F(y)$ increasing.

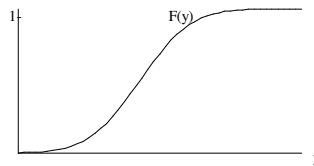


Figure 3:

Probability Density Function (p.d.f.), f ,

Assuming that the derivative exists, we define $f(y) = F'(y)$. See Figure 4.

Properties of f :

- $f(y) \geq 0$
- $\int_{-\infty}^{\infty} f(y) dy = 1$

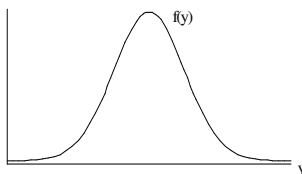


Figure 4:

To Calculate Probabilities for Continuous Random Variables:

$$P(a \leq Y \leq b) = P(Y \leq b) - P(Y < a) = F(b) - F(a).$$

Since $f(y) = F'(y)$, by the Fundamental Theorem of Calculus, we have

$$P(a \leq Y \leq b) = F(b) - F(a) = \int_a^b f(y) dy.$$

See Figure 5.

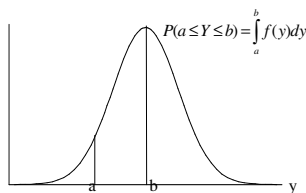


Figure 5:

What does the pdf tell you?

Values of $f(y)$ don't tell you anything directly. If $f(10) = 0.15$, probability that $Y = 10$ is not 0.15. The value of f tells you that the probability that Y is within $10 \leq y \leq 10 + dy$ is $0.15dy$. In other words,

$$P(y \leq Y \leq y + dy) = f(y)dy$$

Thus a large value of $f(y)$ tells you that values near y are very likely. This is why f is called a density function.

Ex: The **uniform distribution** has values of Y between θ_1 and θ_2 which are all equally likely. Thus the p.d.f is

$$f(y) = \begin{cases} \frac{1}{\theta_2 - \theta_1} & \theta_1 \leq y \leq \theta_2 \\ 0 & \text{otherwise.} \end{cases}$$

See Figure 6 for a graph of $f(y)$:

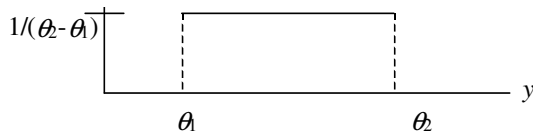


Figure 6: Uniform Distribution

ex. For the uniform distribution with $1 \leq y \leq 6$, find the probability that y is between 3 and 5.

Solution: Since $f(y) = 1/5$, the probability is

$$P(3 < y < 5) = \text{Area of rectangle} = (5 - 3) \frac{1}{5} = \frac{2}{5}.$$

ex: For the pdf in Figure 7, find $P(x > a)$ in terms of a and b .

Solution: We find c in terms of a and b using the fact that the area under the curve is 1. Since

$$\text{Area} = c \left(\frac{a+b}{2} \right) = 1, \text{ we have } c = \frac{2}{a+b}.$$

Thus

$$P(x > a) = \frac{1}{2} c (b - a) = \frac{b - a}{a + b}.$$

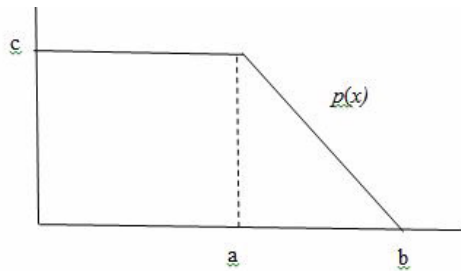


Figure 7: PDF: Find $P(x > a)$

The **exponential distribution** has p.d.f.

$$f(y) = \begin{cases} \frac{1}{\beta} e^{-y/\beta} & \text{for } y \geq 0 \\ 0 & \text{for } y < 0 \end{cases} \quad \text{for } \beta > 0.$$

Its graph is in Figure 8.

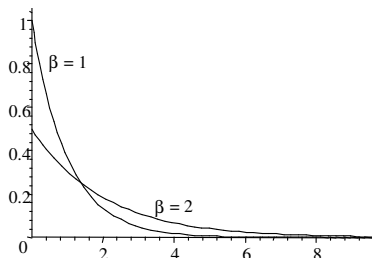


Figure 8: Exponential Distribution

ex. For an exponential distribution, what is the probability of y falling between 0 and 2?

Solution: We integrate to find the area under the curve

$$P(0 \leq Y \leq 2) = \int_0^2 \frac{1}{\beta} e^{-y/\beta} dy = -e^{-y/\beta} \Big|_0^2 = e^0 - e^{-2/\beta} = 1 - e^{-2/\beta}.$$

9.3 Expectation and Variance of Distribution

Population parameters are numbers that characterize a distribution; for example, the expected value, representing the center of the distribution, and the variance, giving the spread of the distribution.

The **Expected Value**, $E(Y)$, of a random variable Y is the the long-term average value, or mean, of Y , weighted by the probabilities of the values.

Discrete:

$$E(Y) = \sum_y yp(y) = \mu$$

Continuous:

$$E(Y) = \int_{-\infty}^{\infty} yf(y)dy = \mu$$

The **Variance**, $V(Y)$, of a random variable gives the spread of the distribution. The variance is the average of the squares of the deviations from the mean; the variance is the square of standard deviation, $\sigma = SD(Y)$. Thus $SD(Y) = \sqrt{V(Y)}$

Discrete:

$$V(Y) = E((Y - \mu)^2) = \sum_y (y - \mu)^2 p(y) = \sigma^2$$

Continuous:

$$V(Y) = E((Y - \mu)^2) = \int_{-\infty}^{\infty} (y - \mu)^2 f(y) dy = \sigma^2$$

We define the **standard deviation**, $SD(Y)$, as

$$SD(Y) = \sqrt{V(Y)}.$$

9.3.1 Examples of Expected Value and Variance

ex. Find $E(Y)$ and $V(Y)$ if Y is the number of heads in three tosses of a coin.

Solution: The p.d.f. and c.d.f. of Y are:

y	0	1	2	3
$p(y)$	1/8	3/8	3/8	1/8
$F(y)$	1/8	4/8	7/8	1

Thus, using the summation formula, we have

$$E(Y) = \sum_y yp(y) = \mu = 0 \cdot \frac{1}{8} + 1 \cdot \frac{3}{8} + 2 \cdot \frac{3}{8} + 3 \cdot \frac{1}{8} = \frac{3}{2}.$$

Notice that the expected value is $3/2$, the mean of 0, 1, 2, 3. In this case the expected value is the mean of the values that Y can take on. This does not always happen; it happens here because the p.d.f of Y is symmetric about the mean.

$$V(Y) = \sum_y (y - \mu)^2 p(y) = \sigma^2 = \left(0 - \frac{3}{2}\right)^2 \cdot \frac{1}{8} + \left(1 - \frac{3}{2}\right)^2 \cdot \frac{3}{8} + \left(2 - \frac{3}{2}\right)^2 \cdot \frac{3}{8} + \left(3 - \frac{3}{2}\right)^2 \cdot \frac{1}{8} = \frac{3}{4}.$$

Thus the standard deviation, $\sigma = \sqrt{3/4} = 0.87$, is roughly the average distance from the mean. Notice that

$$V(Y) = E(Y^2) - \mu^2 = 0^2 \cdot \frac{1}{8} + 1^2 \cdot \frac{3}{8} + 2^2 \cdot \frac{3}{8} + 3^2 \cdot \frac{1}{8} - \left(\frac{3}{2}\right)^2 = 3 - \left(\frac{3}{2}\right)^2 = \frac{3}{4}.$$

ex: For the two pdfs, written $p(x)$, in Figure 9, decide without calculation whether the expected value is more or less than 5. Then find the expected value.

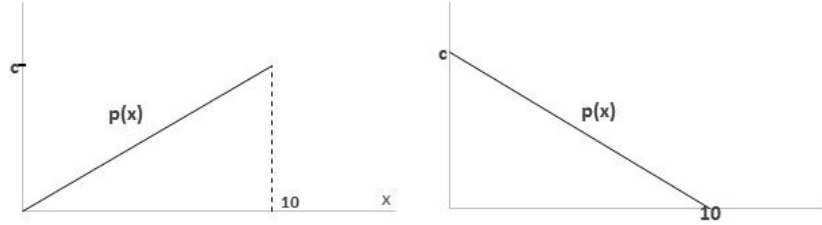


Figure 9: Estimate and calculate Expected Value

Solution: In both pdfs, since the area under the curve is 1, we have

$$\text{Area} = \frac{1}{2} \cdot 10c = 1 \text{ so } c = \frac{1}{5}$$

Left: Because values of more than 5 are more likely than values less than 5, the expected value is greater than 5. Since the slope is $1/50$, the pdf has equation $p(x) = x/50$. Thus

$$EV = \int_0^{10} x \left(\frac{1}{50} x \right) dx = \frac{1}{50} \frac{x^3}{3} \Big|_0^{10} = \frac{20}{3}.$$

Right: Values less than 5 are more likely, so the expected value is less than 5. The equation of the line is $p(x) = 1/5 - x/50$. so

$$EV = \int_0^{10} x \left(\frac{1}{5} - \frac{x}{50} \right) dx = \left(\frac{1}{5} \frac{x^2}{2} - \frac{1}{50} \frac{x^3}{3} \right) \Big|_0^{10} = \frac{10}{3}$$

ex. Calculate the expectation and variance of the exponential distribution

$$f(y) = \begin{cases} \frac{1}{\beta} e^{-y/\beta} & \text{for } y \geq 0 \\ 0 & \text{for } y < 0 \end{cases} \quad \text{for } \beta > 0.$$

Solution: Using the integral formulas and integration by parts, we have

$$\begin{aligned} E(Y) &= \int_0^{\infty} \frac{y}{\beta} e^{-y/\beta} dy = \lim_{k \rightarrow \infty} \left(-y e^{-y/\beta} \Big|_0^k + \int_0^k e^{-y/\beta} dy \right) \\ &= \lim_{k \rightarrow \infty} \left(-y e^{-y/\beta} - \beta e^{-y/\beta} \right) \Big|_0^k = \lim_{k \rightarrow \infty} \left(-k e^{-k/\beta} - \beta e^{-k/\beta} + \beta \right) \\ &= \beta. \end{aligned}$$

$$\begin{aligned}
V(Y) &= E(Y^2) - \beta^2 = \int_0^\infty \frac{y^2}{\beta} e^{-y/\beta} dy - \beta^2 \\
&= \lim_{k \rightarrow \infty} -(y^2 + 2y\beta + 2\beta^2)e^{-y/\beta} \Big|_0^k - \beta^2 = 2\beta^2 - \beta^2 \\
&= \beta^2. \quad \text{So we have } \sigma = \beta.
\end{aligned}$$

9.3.2 General Results about Expectation and Variance:

1. **Theorem:** $V(Y) = E(Y^2) - \mu^2$

Proof: The result is true for discrete and continuous random variables. We prove it for a continuous random variable:

$$\begin{aligned}
V(Y) &= \int_{-\infty}^{\infty} (y - \mu)^2 f(y) dy = \int_{-\infty}^{\infty} (y^2 - 2y\mu + \mu^2) f(y) dy \\
&= \int_{-\infty}^{\infty} y^2 f(y) dy - 2\mu \int_{-\infty}^{\infty} y f(y) dy + \mu^2 \int_{-\infty}^{\infty} f(y) dy \\
&= \int_{-\infty}^{\infty} y^2 f(y) dy - 2\mu^2 + \mu^2 = E(Y^2) - \mu^2.
\end{aligned}$$

2. **Theorem:** $E(aY + b) = aE(Y) + b$

3. **Theorem:** $V(aY + b) = a^2 V(Y)$

Proof:

$$\begin{aligned}
V(aY + b) &= E((aY + b - a\mu - b)^2) = E(a^2(Y - \mu)^2) \\
&= a^2 E((Y - \mu)^2) = a^2 V(Y)
\end{aligned}$$

4. **Theorem:** $E(Y_1 + Y_2) = E(Y_1) + E(Y_2)$ for any random variables Y_1 and Y_2

5. **Theorem:** $V(Y_1 + Y_2) = V(Y_1) + V(Y_2)$ for independent random variables Y_1 and Y_2 .

9.4 Distribution of $g(Y)$ when Distribution of Y is Known

In calculating the variance, we have assumed that the probability of a value of y^2 is the same as the probability of the same value of y . Here we justify and expand this result.

Suppose Y is a continuous random variable with p.d.f $f_Y(y)$. Then $Z = g(Y)$ is another random variable with p.d.f $f_Z(z)$. Suppose g is continuous and monotonic. (If not, divide the domain of g into intervals on which it is continuous and monotonic.)

The relationship between f_Y and f_Z

If dy and dz are related as in Figure 10, then the same events lead to $y \leq Y \leq y + dy$ as to $z \leq Z \leq z + dz$, so

$$P(y \leq Y \leq y + dy) = P(z \leq Z \leq z + dz).$$

Thus, since $P(y \leq Y \leq y + dy) = f_Y(y) dy$, and similarly for Z , we see that

$$f_Y(y) dy = f_Z(z) dz$$

Note: The figure shows a monotonic increasing function, but the result is true when g is decreasing.

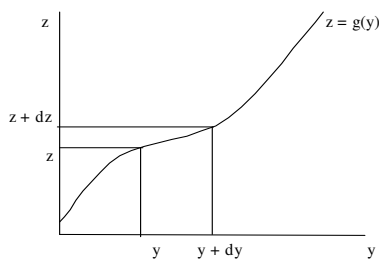


Figure 10:

Calculating $E(g(Y))$

We assume g is monotonic over its whole domain. If not, divide the domain into intervals. Then

$$E(g(Y)) = E(Z) = \int_{-\infty}^{\infty} z f_Z(z) dz = \int_{-\infty}^{\infty} g(y) \cdot f_Z(z) dz = \int_{-\infty}^{\infty} g(y) \cdot f_Y(y) dy,$$

so we have

$$E(g(Y)) = \int_{-\infty}^{\infty} g(y) f_Y(y) dy.$$

This result says that we can use the p.d.f. of Y , that is $f_Y(y)$, to find $E(g(Y))$.

10 TWO RANDOM VARIABLES

10.1 Bivariate Distributions

If we have two random variables X and Y , they may interact or they may be independent of each other. We will analyze their distributions separately and the effect of their interaction (or lack of it) on their sum, $X + Y$.

A **joint distribution** gives the probability of particular values of X and Y occurring together:

$$\text{Joint probability distribution} = p(x, y) = P(X = x, Y = y)$$

$$\text{Joint cumulative distribution} = F(x, y) = P(X \leq x, Y \leq y)$$

A **marginal distribution**, p_1 or p_2 , is the distribution of X or Y alone obtained by summing, or integrating, over the other variable:

$$p_1(x) = \sum_{\text{all } y} p(x, y) \quad \text{and} \quad p_2(y) = \sum_{\text{all } x} p(x, y).$$

A **conditional distribution** is the distribution of one random variable given a specific value of the other one. For example, for each x value we have

$$p(y|x) = \frac{p(x, y)}{p_1(x)}.$$

Independence of random variables is defined in a similar way to independence of events, and has a similar intuitive meaning—that the values of the variables do not affect each other.

Recall that if A, B are independent events, then: $P(A \cap B) = P(A) \cdot P(B)$. If Y_1, Y_2 are **independent random variables**, then the joint distributions are the product of the marginal distributions. In that case,

$$p(x, y) = p_1(x) \cdot p_2(y).$$

If a collection of random variables are independent and each has the same distribution, we say they are **independent, identically distributed**, or **i.i.d.**

When we sample with replacement, each Y_i has the same distribution as the population. If the population is much larger than the sample, then even without replacement, each Y_i has approximately the same distribution as the population. Thus the $Y_1, Y_2, \dots, Y_n$ in a random sample are i.i.d.

10.2 Expectation and Variance of Linear Combinations

1. **Expectation of a Linear Combination:** For any random variables X and Y , we have

$$E(aX + bY) = aE(X) + bE(Y)$$

Interpretation:

This says that the mean of the sum is the sum of the means. Reason: When two random variables are added, each centered around its own mean, the new values center around the sum of the two means. In other words, adding or stretching random variables has the same effect on the mean.

Proof:

Suppose X and Y are discrete random variables. Then:

$$\begin{aligned} E(aX + bY) &= \sum_{\text{all } x, y} (ax + by)p(x, y) \\ &= a \sum_{\text{all } x} x \left(\sum_{\text{all } y} p(x, y) \right) + b \sum_{\text{all } y} y \left(\sum_{\text{all } x} p(x, y) \right) \\ &= a \sum_{\text{all } x} xp_1(x) + b \sum_{\text{all } y} yp_2(y) \\ &= aE(X) + bE(Y) \end{aligned}$$

2. **Variance of a Linear Combination:** If X and Y are independent, then

$$V(aX + bY) = a^2V(X) + b^2V(Y)$$

Interpretation:

This says that the variance of the sum of random variables is the sum of the variances *if* the random variables are independent. Under these circumstances, the variations of the sum of the random variables about the mean is the sum of the variations of each variable about its mean.

If the two random variables are associated (that is, not independent), then the variance of the sum is larger or smaller than the sum of the variances, depending on whether the random variables are positively or negatively associated. The next subsection looks at the covariance which measures the degree and the direction of the association.

Proof:

Suppose the expected values of the two distributions are μ_1 and μ_2 , respectively. Then, the independence of X and Y is used to write the middle term as the product of two factors, each of which turns out to be zero.

$$V(aX + bY) = \sum_{\text{all } x, y} (ax + by - a\mu_1 - b\mu_2)^2 p(x, y)$$

$$\begin{aligned}
&= \sum_{\text{all } x, y} ((ax - a\mu_1) + (by - b\mu_2))^2 p(x, y) \\
&= \sum_{\text{all } x, y} (ax - a\mu_1)^2 p(x, y) + \sum_{\text{all } x, y} 2(ax - a\mu_1)(by - b\mu_2)p(x, y) + \sum_{\text{all } x, y} (by - b\mu_2)^2 p(x, y) \\
&= \sum_{\text{all } x} (ax - a\mu_1)^2 \sum_{\text{all } y} p(x, y) + 2 \sum_{\text{all } x} (ax - a\mu_1)p_1(x) \sum_{\text{all } y} (by - b\mu_2)p_2(y) + \\
&\quad \sum_{\text{all } y} (by - b\mu_2)^2 \sum_{\text{all } x} p(x, y) \\
&= a^2 \sum_{\text{all } x} (x - \mu_1)^2 p_1(x) + 2 \cdot 0 \cdot 0 + b^2 \sum_{\text{all } y} (y - \mu_2)^2 p_2(y) \\
&= a^2 V(X) + b^2 V(Y)
\end{aligned}$$

10.3 Covariance and Correlation

We define the **covariance** of two random variables X and Y , with expected values of μ_1 and μ_2 , as

$$\text{Cov}(X, Y) = \sum_{\text{all } x, y} (x - \mu_1)(y - \mu_2)p(x, y).$$

That is

$$\text{Cov}(X, Y) = E[(X - \mu_1)(Y - \mu_2)]$$

If X and Y have standard deviations σ_X and σ_Y , respectively, we define the **correlation** of X and Y as

$$\text{Correl}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \cdot \sigma_Y}$$

What do covariance and correlation tell us?

- If $\text{cov}(X, Y) > 0$, then X and Y tend to be above their means together and below their means together. In other words, if $\text{cov}(X, Y)$ is positive, then X and Y tend to move together.
- If $\text{cov}(X, Y) < 0$, then when X is above its mean, Y tends to be below its mean, and vice versa. In other words, if $\text{cov}(X, Y)$ is negative, then X and Y tend to move in opposite directions..
- Correlation has no units; Covariance has units which are the product of the units of X and the units of Y .

3. Variance of a Linear Combination: For any random variables X and Y , indendent or not, we have:

$$V(aX + bY) = a^2 V(X) + b^2 V(Y) + 2ab \text{Cov}(X, Y)$$

Interpretation::

- If $\text{cov}(X, Y) > 0$, then $V(aX + bY) > a^2V(X) + b^2V(Y)$.
Reason: X and Y tend to move together, so the variance of their sum is greater than the sum of their variances.
- If $\text{cov}(X, Y) < 0$, then $V(aX + bY) < a^2V(X) + b^2V(Y)$.
Reason: X and Y tend to move in opposite directions, so the variance in their sum is less than the sum of their variances.

Proof:

As before, the expected values of the two distributions are μ_1 and μ_2 . Then:

$$\begin{aligned}
 V(aX + bY) &= \sum_{\text{all } x, y} (ax + by - a\mu_1 - b\mu_2)^2 p(x, y) \\
 &= \sum_{\text{all } x, y} ((ax - a\mu_1) + (by - b\mu_2))^2 p(x, y) \\
 &= \sum_{\text{all } x, y} (ax - a\mu_1)^2 p(x, y) + \sum_{\text{all } x, y} 2(ax - a\mu_1)(by - b\mu_2)p(x, y) + \sum_{\text{all } x, y} (by - b\mu_2)^2 p(x, y) \\
 &= a^2 \sum_{\text{all } x} (x - \mu_1)^2 p_1(x) + 2ab \sum_{\text{all } x, y} (x - \mu_1)(y - \mu_2)p(x, y) + b^2 \sum_{\text{all } y} (y - \mu_2)^2 p_2(y) \\
 &= a^2V(X) + 2ab \text{Cov}(X, Y) + b^2V(Y).
 \end{aligned}$$

4. If X and Y are independent, then $\text{Cov}(X, Y) = 0$.

Proof:

If X, Y are independent $p(x, y) = p_1(x) \cdot p_2(y)$. Thus

$$\text{Cov}(X, Y) = \sum_{\text{all } x, y} (x - \mu_1)(y - \mu_2)p(x, y) = \sum_{\text{all } x} (x - \mu_1)p_1(x) \sum_{\text{all } y} (y - \mu_2)p_2(y) = 0..$$

Note that the converse of this result is not true. $\text{Cov}(X, Y)$ can be 0 without the random variables being independent.

5. For any X and Y , we have $\text{Cov}(X, Y) = E(XY) - \mu_1\mu_2$

Proof:

$$\begin{aligned}
 \text{Cov}(X, Y) &= E[(X - \mu_1)(Y - \mu_2)] \\
 &= E[XY - \mu_1Y - \mu_2X + \mu_1\mu_2] \\
 &= E(XY) - \mu_1E(Y) - \mu_2E(X) + \mu_1\mu_2 \\
 &= E(XY) - \mu_1\mu_2
 \end{aligned}$$

10.4 Examples of Expectation, Variance, and Covariance

An MPAID student is conducting research that compares the incomes of men and women, both of which are normally distributed. She surveys 100 couples (100 men and 100 women) in the same city and finds that incomes of each are normally distributed with the following parameters:

Women:

Mean weekly income	= \$100
Standard deviation	= \$10 (per week)

Men:

Mean weekly income	= \$100
Standard deviation	= \$20 (per week)

Example:

- (a) If both men and women work for 52 weeks a year, what is each one's expected annual income? What is the variance?
- (b) What is the probability that a woman has an income below \$75 per week? What is the probability that a man has an income below this level?

Solution:

- (a) Let X be the random variable giving the weekly income for women and Y be the random variable giving the weekly income for men. Then we want $E(52X)$ and $Var(52X)$ and similarly for Y .

$$\begin{aligned}E(52X) &= 52 \cdot E(X) = 52 \cdot \$100 = \$5,200 \\E(52Y) &= 52 \cdot E(Y) = 52 \cdot \$100 = \$5,200 \\Var(52X) &= 52^2 Var(X) = 52^2 (10)^2 = 270,400 \text{ dollars}^2 \\Var(52Y) &= 52^2 Var(Y) = 52^2 (20)^2 = 1,081,600 \text{ dollars}^2\end{aligned}$$

- (b) Using the standard normal variable Z , we have

$$\begin{aligned}P(X \leq 75) &= P(Z \leq ((75 - 100)/10)) = P(Z \leq -2.5) = 0.0062 = 0.62\% \\P(Y \leq 75) &= P(Z \leq ((100 - 75)/20)) = P(Z \leq -1.25) = 0.1056 = 10.56\%\end{aligned}$$

Example continued:

Now the student wants to describe the joint incomes of couples.

- (c) What is the expected joint weekly income of a couple consisting of a man and a woman, both of whom are working?
- (d) What is the expected variance on this joint weekly income, if there is no relationship between the man's and woman's income (that is, the correlation of their incomes is zero?)
- (e) Now suppose that there is selective mating within couples, so that high-income men are more likely to marry high-income women and vice versa. If the correlation between the separate incomes is 100%, what is the expected variance of the joint income?
- (f) What does a negative covariance mean in this context?

Solution: (c) The random variable giving the joint weekly income is $X + Y$, and

$$E(X + Y) = E(X) + E(Y) = 100 + 100 = \$200$$

(d) We have

$$\begin{aligned} \text{Var}(X + Y) &= \text{Var}(X) + \text{Var}(Y) + 2\text{Covar}(X, Y) \\ &= 10^2 + 20^2 \\ &= 500 \text{ dollars}^2 \end{aligned}$$

(e) Since $\text{Cov}(X, Y) = \text{Correl}(X, Y) \cdot \text{Sd}(X) \cdot \text{Sd}(Y)$, we have

$$\begin{aligned} \text{Var}(X + Y) &= \text{Var}(X) + \text{Var}(Y) + 2\text{Covar}(X, Y) \\ &= 10^2 + 20^2 + 2 \cdot 100\% \cdot 10 \cdot 20 \\ &= 900 \text{ dollars}^2 \end{aligned}$$

(f) A negative covariance means that income tends to vary more within couples than between couples. In other words, high-earning women tend to live with low-earning men and vice versa. This might reflect a choice of couples to specialize, with one working longer hours and the other looking after children or a home.